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Michael Baker

LOG OF BORING NO.: MB-5

INTERNATIONAL PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services

SHEET: 2 OF 4

LATITUDE: LONGITUDE: GEO. DATUM: NAD83

START: 1/21/2024 END: 1/25/2023

NORTH: 5239049.933

EAST: 726689.5856

COORD. DATUM: UTM21-North

LOGGER: Dylan Brown

STATION: OFFSET: BASELINE:

DRILLER: Brendan Clark

EQUIPMENT: CME 55 Marooka

DRILL. CO.: Logan Geotech

METHOD DETAILS: HQ series core, w/ water, 1.375 in ID spoon. Automatic hammer; 140 lb; 30 inch fall; ER = 80% (assumed).

▼ H₂O @ 19.9 C. CAVED @ TIME: 15:00 DATE: 1/25/2024 DESC.: 0 Hr. PLUNGE: -90

▼ H₂O @ 19.9 C. CAVED @ TIME: 08:26 DATE: 1/26/2024 DESC.: 17.5 Hr. ELEV. DATUM: CGVD2013

▼ H₂O @ C. CAVED @ TIME: DATE: DESC.: GROUND ELEV.: 24

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	BLOWS/0.5 FT or (ROD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
21.0	A-N			BAG 2(20.0-71.9) PL=NP MC=22.1 USCS=GP-GM P#200=7.1		20		
22.5	S-15	40%	22			21		
	A-N					22		
26.0	S-16	100%	14	S-16(26.0-27.5) PL=NP MC=22.1 USCS=SP-SM P#200=9.2		23		
27.5	A-N					24		
31.0	S-17	0%	50/0.3			25		
36.0	S-18	67%	16			26		
37.5	A-N					27		
40.0	A-N					28		
41.0	A-N					29		
42.5	A-N					30		
44.0	A-N					31		
46.0	A-N					32		
47.5	A-N					33		
51.0	S-21	67%	29			34		
52.5	A-N					35		
56.0	S-22	67%	48			36		
57.0	A-N					37		
60.0	A-N					38		
61.0	A-N					39		
62.5	A-N					40		
64.0	A-N					41		
66.0	A-N					42		
67.5	A-N					43		
71.0	A-N					44		
71.9	A-N					45		
71.9	A-N					46		
71.9	A-N					47		
71.9	A-N					48		
71.9	A-N					49		
71.9	A-N					50		
71.9	A-N					51		
71.9	A-N					52		
71.9	A-N					53		
71.9	A-N					54		
71.9	A-N					55		
71.9	A-N					56		
71.9	A-N					57		
71.9	A-N					58		
71.9	A-N					59		
71.9	A-N					60		

Boring terminated at 71.9' due to tool loss downhole and homogeneity of the material encountered. Boring backfilled cement bentonite grout.

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LOG OF BORING NO.: MB-5

INTERNATIONAL PROJECT: Mulpoint Joint Maritime Facility SMIG PRI Engineering Services

SHEET: 3 OF 4

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▼ H₂O @ C. CAVED @ TIME: DATE: DESC.: GROUND ELEV.: 24

SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	BLOWS/0.5 FT or (ROD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
41.0	S-19	73%	38	S-19(S-20/S-21(41.0-52.5) PL=NP MC=8.2 USCS=SP-SM P#200=9.5		40		
42.5	A-N					41		
46.0	S-20	100%	42			42		
47.5	A-N					43		
51.0	S-21	67%	29			44		
52.5	A-N					45		
56.0	S-22	67%	48			46		
57.0	A-N					47		
60.0	A-N					48		
61.0	A-N					49		
62.5	A-N					50		
64.0	A-N					51		
66.0	A-N					52		
67.5	A-N					53		
71.0	A-N					54		
71.9	A-N					55		
71.9	A-N					56		
71.9	A-N					57		
71.9	A-N					58		
71.9	A-N					59		
71.9	A-N					60		

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SAMPLE DEPTH (FT)	SAMPLE TYPE - NUMBER	RECOVERY %	BLOWS/0.5 FT or (ROD)	TEST RESULTS	STRATA	DEPTH	DESCRIPTION	REMARKS
61.0	S-23	53%	35			60		
62.5	A-N					61		
65.0	S-24	100%	36			62		
66.0	A-N					63		
71.0	S-25	56%	45			64		
71.9	A-N					65		
71.9	A-N					66		
71.9	A-N					67		
71.9	A-N					68		
71.9	A-N					69		
71.9	A-N					70		
71.9	A-N					71		
71.9	A-N					72		
71.9	A-N					73		
71.9	A-N					74		
71.9	A-N					75		
71.9	A-N					76		
71.9	A-N					77		
71.9	A-N					78		
71.9	A-N					79		
71.9	A-N					80		

Boring terminated at 71.9' due to tool loss downhole and homogeneity of the material encountered. Boring backfilled cement bentonite grout.

NOTES

1. 1 METER = 3.28 FEET



APPROVED: Digitally signed by signature
Win B Forbes
Date: 2026.01.28
16:54:04-05'00'
FOR COMMANDER NAVFAC

ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026
DES LAP DRW LAP CHG CFC
PM/DM MJD/ATP
BRANCH MANAGER
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
SUBSURFACE EXPLORATION RECORDS

SCALE: AS NOTED
EPROOF NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157078
SHEET 16 OF 170
B-606

FILE NAME: Y:\OCONUS\MultiplateSheets\2026_1783264-P1360_IAMF\B_Design\DWG\Sheet\04_GEO\1783264-B-601.dwg LAYOUT NAME: B-607 PLOTTED: Wednesday, January 21, 2026 - 8:07am USER: brampton

TABLE 1
SUMMARY OF SUBSURFACE EXPLORATION
MULPOINT JOINT MARTIME FACILITY SMIG PRI ENGINEERING SERVICES
ARGENTIA, NEWFOUNDLAND AND LABRADOR, CANADA

BORING NO.	Ground Surface Elevation	SOIL										ROCK		Bottom of Boring Elevation	0-Hour Water Elevation	24-Hour Water Elevation	
		Unsampled	Pavement/S ubbase	Fill/ Topsoil	Average N-Value	Tidal	Average N-Value	Glacial	Average N-Value	Glacio-Marine	Average N-Value	Soil Sampled	Top of Rock Elevation				Rock Cored
		(ft-msl)	(ft)	(ft)	(SPT N)	(ft)	(SPT N)	(ft)	(SPT N)	(ft)	(SPT N)	(ft)	(ft)				(ft-msl)
MB-1	14.9	--	--	6.0	23	--	--	-	--	--	--	6.0	--	--	8.9	--	--
MB-1A	14.9	--	--	12.0	17	--	--	68.1	>100	20.0	57	100.1	--	--	-85.2	3.9	3.5
MB-1B	14.9	10.0	--	--	--	--	--	--	--	--	--	--	--	--	4.9	Dry	--
MB-2	14.2	--	--	12.0	28	--	--	69.5	>100	18.8	40	100.3	--	--	-86.1	4.3	3.9
MB-2A	14.2	10.0	--	--	--	--	--	--	--	--	--	--	--	--	4.2	Dry	--
MB-3	11.4	--	0.2	7.3	40	1.5	7	21.0	40	--	--	30.0	--	--	-18.6	--	--
MB-3A	11.4	30.0	--	--	--	--	--	53.6	>100	5.0	64	70.1	--	--	-88.7	1.4	2.0
MB-4	12.4	--	--	3.0	38	--	--	67.1	80	30.0	70	100.1	--	--	-87.7	0.0	0.3
MB-5	24.0	--	--	0.4	--	--	--	71.5	80	--	--	71.9	--	--	-47.9	4.1	4.1

TABLE 2
SUMMARY OF SOIL LABORATORY TESTING
MULPOINT JOINT MARTIME FACILITY SMIG PRI ENGINEERING SERVICES
ARGENTIA, NEWFOUNDLAND AND LABRADOR, CANADA

BORING NO.	SAMPLE NO.	SAMPLE DEPTH INTERVAL (feet)	MOISTURE CONTENT (%)	ATTERBERG LIMITS			USCS GRADATION				USCS CLASS.	SOIL CORROSIIVITY PARAMETERS				WATER CORROSIIVITY PARAMETERS			
				LL	PL	PI	GRAVEL (%)	SAND (%)	SILT (%)	CLAY (%)		pH (s.u.)	Chloride (mg/kg)	Sulfate (mg/kg)	Resistivity (kiloohm*c m)	pH (s.u.)	Chloride (mg/L)	Sulfate (mg/L)	Conductivity (µS/cm)
MB-1A	S-9 to S-13	12.0-19.5	6.8	NP	NP	NP	62.6	29.5	5.1	2.8	GP-GM	--	--	--	--	--	--	--	--
MB-1A	S-15/S-16	26.0-32.5	10.8	NP	NP	NP	22.2	65.2	7.1	5.5	SP-SM	--	--	--	--	--	--	--	--
MB-1A	S-19 to S-21	46.0-57.5	19.7	NP	NP	NP	0.6	87.7	8.4	3.3	SP-SM	--	--	--	--	--	--	--	--
MB-1A	S-22	61.0-62.5	13.6	22	13	9	0.0	17.8	51.9	30.3	CL	--	--	--	--	--	--	--	--
MB-1A	S-23	66.0-67.5	16.8									--	--	--	--	--	--	--	--
MB-1A	S-24	71.0-72.5	14									--	--	--	--	--	--	--	--
MB-1A	S-25	76.0-77.5	16.5									--	--	--	--	--	--	--	--
MB-1B	BAG 1	0.0-10.0	16.5	23	22	1	38.7	44.7	12.1	4.4	SM	8.59	15	33	5.146	--	--	--	--
MB-1A	BAG 1	21.0-61.0	17.8	NP	NP	NP	7.5	82.8	7.0	2.7	SP-SM	8.41	<5	48	9.313	--	--	--	--
MB-1A	Jug 1	--	--	--	--	--	--	--	--	--	--	--	--	--	--	7.75	24.3	3.9	161
MB-2	S-2 to S-6	1.5-9.0	9.7	NP	NP	NP	52.3	38.3	6.4	3.1	GP-GM	--	--	--	--	--	--	--	--
MB-2	S-11 to S-14	15.0-21.0	7.6	NP	NP	NP	57.7	35.1	5.9	1.4	GP-GM	--	--	--	--	--	--	--	--
MB-2	S-25/S-26	76.0-82.5	16.4	21	15	6	0.1	21.0	50.5	28.4	CL-ML	--	--	--	--	--	--	--	--
MB-2	S-27	86.0-87.5	22.8	36	20	16	0	3.1	40.3	56.7	CL	--	--	--	--	--	--	--	--
MB-2A	Bag 1	0.0-10.0	15.6	24	19	5	34.1	44.5	16.2	5.2	SC-SM	7.26	15	60	7.973	--	--	--	--
MB-2	Bag 2	15.0-71.0	17.8	NP	NP	NP	7.8	74.6	13.4	4.2	SM	8.85	24	21	4.556	--	--	--	--
MB-2	Jug 1	--	--	--	--	--	--	--	--	--	--	--	--	--	--	7.64	21.2	2.49	136
MB-3	S-2 to S-4	1.5-6.0	5.3	15	12	3	41.2	35.4	15.2	8.2	SM	--	--	--	--	--	--	--	--
MB-3	S-6	7.5-9.0	75.6	NP	NP	NP	32.3	46.9	15.4	5.4	SM	--	--	--	--	--	--	--	--
MB-3	S-9 to S-11	12.0-16.5	8.1	NP	NP	NP	55.5	20.6	18.6	5.3	GM	--	--	--	--	--	--	--	--
MB-3	S-14/S-15	19.5-22.5	12.2	NP	NP	NP	25.5	33.4	33.2	8.0	SM	--	--	--	--	--	--	--	--
MB-3	Bag 1	0.2-10.0	17.7	NP	NP	NP	32.2	51.7	12.0	4.1	SM	7.52	18	51	5.461	--	--	--	--
MB-3A	S-2 to S-5	31.5-36.9	7.0	NP	NP	NP	40.4	46.6	8.3	4.7	SM	--	--	--	--	--	--	--	--
MB-3A	S-18 to S-21	55.5-60.0	20.6	NP	NP	NP	21.2	63.5	11.9	3.3	SM	--	--	--	--	--	--	--	--

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SUMMARY OF SOIL LABORATORY TESTING
MULPOINT JOINT MARTIME FACILITY SMIG PRI ENGINEERING SERVICES
ARGENTIA, NEWFOUNDLAND AND LABRADOR, CANADA

BORING NO.	SAMPLE NO.	SAMPLE DEPTH INTERVAL (feet)	MOISTURE CONTENT (%)	ATTERBERG LIMITS			USCS GRADATION				USCS CLASS.	SOIL CORROSIIVITY PARAMETERS				WATER CORROSIIVITY PARAMETERS			
				LL	PL	PI	GRAVEL (%)	SAND (%)	SILT (%)	CLAY (%)		pH (s.u.)	Chloride (mg/kg)	Sulfate (mg/kg)	Resistivity (kiloohm*c m)	pH (s.u.)	Chloride (mg/L)	Sulfate (mg/L)	Conductivity (µS/cm)
MB-3A	S-25	67.5-69.0	27.2	26	15	11	1.2	10.3	50.3	38.2	CL	--	--	--	--	--	--	--	--
MB-3A	S-26	69.0-70.5	16.8									--	--	--	--	--	--	--	--
MB-3A	S-27	70.5-72.0	15.9									--	--	--	--	--	--	--	--
MB-3A	S-32 to S-34	76.5-79.5	19.3	21	14	7	0.0	14.7	56.1	29.2	CL	--	--	--	--	--	--	--	--
MB-3A	S-37/S-38	90.0-95.9	8.4	13	11	2	39.1	38.1	17.1	5.8	SM	--	--	--	--	--	--	--	--
MB-3A	Bag 1	30.0-100.1	17.9	NP	NP	NP	8.3	83.4	5.3	3.0	SP-SM	8.29	<5	15	8.710	--	--	--	--
MB-3A	Jug 1	--	--	--	--	--	--	--	--	--	--	--	--	--	--	7.35	37.3	5.70	210
MB-4	Bag 1	0.0-12.0	10.0	NP	NP	NP	34.7	47.4	13.3	4.6	SM	8.42	<5	27	7.102	--	--	--	--
MB-4	Bag 2	12.0-82.5	18.9	NP	NP	NP	6.9	66.8	19.5	6.7	SM	8.52	<5	51	6.358	--	--	--	--
MB-4	S-3 to S-8	3.0-12.0	8.5	15	14	1	34.9	44.7	15.4	5.0	SM	--	--	--	--	--	--	--	--
MB-4	S-14 to S-16	19.5-32.5	19.8	NP	NP	NP	2.6	83.2	10.9	3.2	SM	--	--	--	--	--	--	--	--
MB-4	S-21/S-22	56.0-62.5	19.2	15	13	2	0.0	55.6	28.3	16.1	SM	--	--	--	--	--	--	--	--
MB-4	S-23	66.0-67.5	15.0	18	13	5	1.1	16.3	61.1	21.5	CL-ML	--	--	--	--	--	--	--	--
MB-4	S-24	71.0-72.5	15.4	24	14	10	0.0	20.3	48.5	31.2	CL	--	--	--	--	--	--	--	--
MB-4	S-25	76.0-77.5	19.8									--	--	--	--	--	--	--	--
MB-4	S-26 to S-28	81.0-92.5	12.1									--	--	--	--	--	--	--	--
MB-4	Jug 1	--	--	--	--	--	--	--	--	--	--	--	--	--	--	7.66	23.8	3.07	149
MB-5	Bag 1	0.0-18.0	13.0	NP	NP	NP	47.8	44.6	4.8	2.7	SP-SM	8.12	<5	24	7.973	--	--	--	--
MB-5	Bag 2	20.0-71.9	22.1	NP	NP	NP	55.0	37.8	5.1	2.0	GP-GM	8.56	24	21	6.338	--	--	--	--
MB-5	S-5 to S-7	6.0-10.5	10.2	17	11	6	47.1	43.8	5.4	3.7	GP-GM	--	--	--	--	--	--	--	--
MB-5	S-16	26.0-27.5	22.1	NP	NP	NP	19.8	71.0	4.8	4.3	SP-SM	--	--	--	--	--	--	--	--
MB-5	S-19 to S-21	41.0-52.5	8.2	NP	NP	NP	42.4	48.0	5.5	4.1	SP-SM	--	--	--	--	--	--	--	--
MB-5	Jug 1	--	--	--	--	--	--	--	--	--	--	--	--	--	--	7.35	25.3	3.00	150

TABLE 3
ENVIRONMENTAL LABORATORY ANALYTICAL RESULTS FOR CONSTITUENTS DETECTED IN SOIL
MULPOINT JOINT MARTIME FACILITY SMIG PRI ENGINEERING SERVICES
ARGENTIA, NEWFOUNDLAND AND LABRADOR, CANADA

	Sample Designation	Atlantic RBCA	USEPA	24020503 MB-1B	24020504 MB-1B	24020501 MB-2A	24020502 MB-2A	24011001 MB-3	24012601 MB-4	24012201 MB-5
	Sample Depth	Industrial	Industrial	4.0' - 7.0'	7.0' - 10.0'	4.0' - 7.0'	7.0' - 10.0'	7.5' - 9.0'	9.0' - 10.5'	18.0' - 19.5'
	Date Sampled	EQS ⁽¹⁾	RSLs ⁽²⁾	2/5/2024	2/5/2024	2/5/2024	2/5/2024	1/10/2024	1/26/2024	1/22/2024
Atlantic RBCA Tier 1 Hydrocarbons (mg/kg)										
Xylenes (Total)	60	250	0.07		0.05 U	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
C6-C10 (less BTX)	NA	NE	7		3 U	3 U	3 U	3 U	3 U	3 U
>C10-C16 Hydrocarbons	NA	NE	969		1,220	15 U	15 U	15 U	15 U	15 U
>C16-C21 Hydrocarbons	NA	NE	3,600		4,680	25	15 U	18	15 U	15 U
>C21-C32 Hydrocarbons	NA	NE	5,060		7,000	539	186	151	15 U	15 U
Modified TPH (Tier 1)	10,000	NE	9,640		12,900	564	186	169	15 U	15 U
Polycyclic Aromatic Hydrocarbons (mg/kg)										
1-Methylnaphthalene	560	73	19.0		21.5	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
2-Methylnaphthalene	560	300	35.2		39.3	0.01 U	0.01 U	0.01 U	0.01 U	0.01 U
Acenaphthene	43,000	4,500	51.3		87.4	0.0599	0.0396	0.0076	0.00671 U	0.00671 U
Acenaphthylene	66	4,500 ⁽⁴⁾	1.34		1.65	0.004 U	0.004 U	0.004 U	0.004 U	0.004 U
Acridine	NE	NE	254		497	0.05 U	0.05 U	0.05 U	0.05 U	0.05 U
Anthracene	300,000	23,000	120		234	0.13	0.09	0.03 U	0.03 U	0.03 U
Benzo(e)pyrene	NE	7.3	54.3		103	0.10	0.06	0.05 U	0.05 U	0.05 U
Fluoranthene	50,000	3,000	297		568	0.52	0.32	0.05 U	0.05 U	0.05 U
Fluorene	39,000	3,000	86.0		146	0.08	0.05	0.01 U	0.01 U	0.01 U
Naphthalene	25	8.6	101		91.8	0.01 U	0.01 U	0.03	0.01 U	0.01 U
Perylene	NE	6.7	20.3		29.4	0.05 U	0.05 U	0.38	0.05 U	0.05 U
Phenanthrene	30,000 ⁽⁴⁾	2,300 ⁽⁴⁾	399		772	0.42	0.27	0.05	0.03 U	0.03 U
Pyrene	30,000	2,300	223		424	0.40	0.24	0.05 U	0.05 U	0.05 U
Quinoline	NE	0.77	1,850		4,200	0.10	0.06	0.05 U	0.05 U	0.05 U
BaP Total Potency Equivalent ⁽⁵⁾	5.3	NA	141		258	0.21	0.13	0.01	0.01	0.01
Benzo(a)anthracene	NA	21	132		241	0.19	0.12	0.01 U	0.01 U	0.01 U
Benzo(a)pyrene	NA	2.1	91.1		158	0.14	0.09	0.01 U	0.01 U	0.01 U
Benzo(b)fluoranthene	NA	21	150		400	0.24	0.09	0.05 U	0.05 U	0.05 U
Benzo(j+k)fluoranthene	NA	211.8 ⁽⁵⁾	72.7		128	0.13	0.09	0.05 U	0.05 U	0.05 U
Benzo(ghi)perylene	NA	2,300 ⁽⁴⁾	36.9		56.8	0.08	0.05	0.01 U	0.01 U	0.01 U
Chrysene	NA	2,100	120		223	0.20	0.11	0.01 U	0.01 U	0.01 U
Dibenzo(a,h)anthracene	NA	2.1	8.58		12.3	0.006 U	0.006 U	0.006 U	0.006 U	0.006 U
Indeno(1,2,3)pyrene	NA	21	44.9		81.9	0.07	0.06	0.01 U	0.01 U	0.01 U
Volatile Organic Compounds (mg/kg)										
Benzene	0.52	5.1	0.0068 U		0.008	0.0068 U	0.0068 U	0.0068 U	0.0068 U	0.0068 U
Toluene	4,700	4,700	0.1		0.08 U	0.08 U	0.08 U	0.08 U	0.08 U	0.08 U
Ethylbenzene	10,000	25	0.208		0.121	0.018 U	0.018 U	0.018 U	0.018 U	0.018 U
Xylenes (Total)	60	250	0.7		0.3	0.1 U	0.1 U	0.1 U	0.1 U	0.1

GENERAL NOTES

- THIS SURVEY DOES NOT GUARANTEE THE EXISTENCE OR NONEXISTENCE, SIZE, TYPE, DEPTH, MATERIAL OR LOCATION OF ANY UNDERGROUND UTILITIES. ALL UTILITIES SHOWN ARE BASED ON ABOVEGROUND UTILITY STRUCTURES, AVAILABLE UTILITY MAPS AND SITE PLANS. THERE MAY BE OTHERS, THE EXISTENCE OF WHICH IS NOT PRESENTLY KNOWN. VERIFY ALL EXISTING SITE UTILITIES PRIOR TO ANY EXCAVATION. VERIFY IN THE FIELD THE EXACT LOCATION OF THE POINTS OF CONNECTION BETWEEN NEW AND EXISTING UTILITIES.
2. ENSURE COMPLIANCE WITH SECURITY REGULATIONS AND ACCESS REQUIREMENTS FOR CONTRACTOR PERSONNEL ENTERING THE PROJECT SITE. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL BADGING AND TRAINING REQUIRED FOR BASE AND RESTRICTED AREA ACCESS.
3. CONTRACTOR VEHICLES MUST ABIDE BY BASE ROADWAY REGULATIONS. COMMERCIAL VEHICLES AND HEAVY EQUIPMENT MUST TRAVEL ALONG PREDETERMINED HAUL ROUTES AND PASS THROUGH SECURITY ACCESS POINTS.
4. BRIEF PERSONNEL ON APPROVED HAUL ROUTES, RESTRICTED AREAS, SMOKING POLICY AND CELL PHONE POLICY.
5. COORDINATE EQUIPMENT LAYDOWN AREAS AND HAUL ROUTES WITH THE CONTRACTING OFFICER. OBTAIN APPROVAL FROM THE CONTRACTING OFFICER PRIOR TO COMMENCEMENT OF CONSTRUCTION ACTIVITIES. THE CONTRACTOR MUST BE RESPONSIBLE FOR PROVIDING TEMPORARY FENCING AND SECURING THESE AREAS.
6. SUBMIT A TRAFFIC CONTROL PLAN TO THE CONTRACTING OFFICER FOR APPROVAL. ALL ROADS MUST BE OPEN TO TRAFFIC. TRAFFIC CONTROL PLAN SHALL MEET THE REQUIREMENTS OF THE MUTCD (MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES, UNITED STATES FEDERAL HIGHWAY ADMINISTRATION) AND HOST NATION REQUIREMENTS.
7. REPAIR PAVEMENTS DAMAGED DURING CONSTRUCTION TO MATCH EXISTING PAVEMENT.
8. ADJUST TOP OF EXISTING UTILITY STRUCTURES AS REQUIRED TO BE FLUSH WITH THE FINISHED GRADE ELEVATION.
9. MATCH THE ELEVATIONS OF NEW PAVEMENTS TO THE EXISTING ADJACENT PAVEMENT ELEVATIONS UNLESS NOTED OTHERWISE.
10. DIMENSIONS AND ELEVATIONS ARE IN METERS UNLESS OTHERWISE INDICATED. PIPE DIAMETERS ARE IN MILLIMETERS AND REFER TO INSIDE DIAMETER.
11. DIMENSIONS SHOWN ARE TO FACE OF BUILDING, EDGE OF PAVEMENT AND FACE OF CURB UNLESS OTHERWISE INDICATED.
12. REMOVE SOLID WASTE MATERIALS GENERATED DURING DEMOLITION ACTIVITIES THAT CANNOT BE REUSED OR RECYCLED FROM GOVERNMENT PROPERTY AND DISPOSE OF IN AN APPROVED LANDFILL SITE. IF HAZARDOUS MATERIALS ARE ENCOUNTERED DURING DEMOLITION ACTIVITIES, IMMEDIATELY CONTACT THE CONTRACTING OFFICER.
13. PARKING SPACES PROVIDED:
 - 1 REGULAR
 - 1 VAN ACCESSIBLE
 - 2 TOTAL
14. UNLESS OTHERWISE INDICATED, ALL EXISTING STRUCTURES AND FACILITIES MUST REMAIN.
15. NO EARTH WORK/LAND DISTURBANCE IS TO BE PERFORMED UNTIL ALL SOIL EROSION AND SEDIMENT CONTROL DEVICES ARE IN PLACE.
16. ALL GRADE TRANSITIONS BETWEEN NEW AND EXISTING MUST BE SMOOTH AND GRADUAL WITH NO SHARP OR ABRUPT CHANGES.
17. COORDINATE THE WORK OF ALL TRADES – VERIFY ALL FIELD CONDITIONS, QUANTITIES AND DIMENSIONS PRIOR TO COMMENCEMENT OF WORK. ANY DISCREPANCIES MUST BE PROMPTLY BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER.
18. EXECUTE ALL WORK WITH CARE TO PROTECT FROM DAMAGE ALL ADJACENT EXISTING FEATURES THAT ARE TO REMAIN. ANY SUCH DAMAGE MUST BE REPAIRED OR REPLACED TO MATCH THE ORIGINAL CONDITION AS APPROVED BY THE CONTRACTING OFFICER AT NO ADDITIONAL COST TO THE GOVERNMENT.
19. FINISHED GRADE OF TOP SOIL (AFTER COMPACTION) MUST BE 0.5" TO 1" (MAXIMUM) BELOW TOP OF ABUTTING PAVEMENTS, SIDEWALKS AND CURBING.
20. EQUIPMENT AND TRUCK TRACK-TIRE CLEANING/FLUSHING STATIONS MUST BE ESTABLISHED AS NECESSARY TO ENSURE THAT NO MUD OR DEBRIS MIGRATES FROM THE CONSTRUCTION SITE TO THE BASE ROADS.
21. ALL HOLES, DEPRESSIONS, EXCAVATIONS, ETC. CREATED AS A RESULT OF DEMOLITION MUST BE BACK-FILLED, COMPACTED, GRADED AND SEEDDED.
22. FIELD VERIFY THE ELEVATIONS OF ALL CONNECTIONS TO EXISTING UTILITIES PRIOR TO CONSTRUCTING UPSTREAM PORTIONS OF THE NEW UTILITIES. ANY DISCREPANCIES MUST PROMPTLY BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER.
23. IF AN UNMARKED WELL IS DISCOVERED AND INADVERTENTLY DAMAGED, STOP WORK AND IMMEDIATELY CONTACT THE CONTRACTING OFFICER FOR FURTHER INSTRUCTIONS.

SURVEY NOTES

1. COORDINATES ARE BASED ON THE UNIVERSAL TRANSVERSE MERCATOR COORDINATE SYSTEM, ZONE 21 NORTH (UTM21N), NORTH AMERICAN DATUM OF 1983 (NAD83) HORIZONTAL DATUM AND THE CANADIAN GEODETIC VERTICAL DATUM OF 2013 (CGVD2013a).
2. ALL UNITS FOR THIS SITE ARE IN METERS UNLESS OTHERWISE NOTED.
3. A SURVEY WAS COMPLETED BY MICHAEL BAKER INTERNATIONAL USING A COMBINATION OF STATIC LIDAR SCANNING AND CONVENTIONAL GPS.
 - A. THE TOPOGRAPHIC SURVEY WAS COMPLETED ON NOVEMBER 16, 2023. THE UNDERGROUND UTILITY AND GEOTECH BORING SURVEY WAS COMPLETED ON FEBRUARY 1, 2024.
4. UNDERGROUND UTILITIES WERE LOCATED BY ALLNORTH USING GROUND PENETRATING RADAR (GPR).
5. ALL UNDERGROUND UTILITIES SHOWN ARE APPROXIMATE AND HAVE BEEN COMPILED USING GPR DATA MENTIONED ABOVE AND A SITE PLAN TITLED "TERMINAL EQUIPMENT BUILDING ADDITION" PREPARED BY McGAUGHY, MARSHALL & McMILLAN ARCHITECTS AND ENGINEERS DATED APRIL 10TH, 1979.
6. A FIELD INVESTIGATION SITE VISIT WAS COMPLETED BY NAVFAC LANT ON JUNE 3, 2025. THE SURVEY HAS BEEN SUPPLEMENTED WITH ADDITIONAL HISTORIC UTILITY INFORMATION AND INFORMATION GATHERED FROM THE SITE VISIT.

 SURVEY CONTROL				
POINT No.	NORTHING	EASTING	ELEVATION	DESCRIPTION
* 100	5239335.211	726704.154	7.333	REBAR W/CAP
* 101	5239340.945	5239340.945	6.691	REBAR W/CAP
102	5239207.390	726617.866	3.062	REBAR W/CAP
103	5239068.610	726735.345	9.665	REBAR W/CAP

CONTROL WAS SET BY MICHAEL BAKER INTERNATIONAL ON THE FIRST DAY OF SURVEY.

ALL REBAR W/CAP ARE 750mm IN LENGTH AND IDENTIFIED BY A PLASTIC CAP BEARING THE IMPRINT
"MICHAEL BAKER INTERNATIONAL"

ABBREVIATIONS

ABA	AMERICAN BARRIERS ACT
ALUM	ALUMINUM
APPROX	APPROXIMATE
BM	BENCHMARK
CB	CATCH BASIN
CIP	CAST IRON PIPE
CL	CENTERLINE
CMP	CORRUGATED METAL PIPE
CO	CLEAN OUT
COMM(S)	COMMUNICATIONS
CONC	CONCRETE
DEPT	DEPARTMENT
DI	DROP INLET
Ø, DIA	DIAMETER
DIM	DIMENSION
DIP	DUCTILE IRON PIPE
DOR	DESIGNER OF RECORD
DWGS	DRAWINGS
ELEC	ELECTRICAL
ELEV, EL	ELEVATION
EP	EDGE OF PAVEMENT
EQ	EQUAL
EXIST, EX	EXISTING
FDN	FOUNDATION
FED	FEDERAL
FFE	FINISH FLOOR ELEVATION
FG	FINISH GRADE
FO	FIBER OPTIC
FH	FIRE HYDRANT
FW	FIRE WATER
GALV	GALVANIZED
H/C	HANDICAP
HORIZ	HORIZONTAL
HW	STORM SEWER HEADWALL STRUCTURE
INV	INVERT
KO	CONTRACTING OFFICER
KTR	CONTRACTOR
M	METER
LOC	LIMITS OF CONSTRUCTION
LONG	LONGITUDINAL
LP	LIGHT POLE
MAX	MAXIMUM
MC	MATCH EXISTING GRADE
MIN	MINIMUM
MFR	MANUFACTURER
MH	MANHOLE
MM	MILLIMETER
NTS	NOT TO SCALE
OC	ON CENTER
OD	OUTSIDE DIAMETER
OWS	OIL/WATER SEPARATOR
PERF	PERFORATED
PP	POWER POLE
PVC	POLYVINYL CHLORIDE
PVT	PAVEMENT
R	RADIUS
RCP	REINFORCED CONCRETE PIPE
RD	ROAD DRAIN, ROAD
REINF	REINFORCED, -MENT
REQ'D	REQUIRED
SS, SAN	SANITARY SEWER
SCHED	SCHEDULE
SD	STORM DRAIN
SFM	SANITARY FORCE MAIN
SHT	SHEET
SMH	SEWER MANHOLE
SPEC	SPECIFICATION
SQ	SQUARE
S/W	SIDEWALK
TBM	TEMPORARY BENCHMARK
TEL	TELEPHONE
TEMP	TEMPORARY
TG	TOP OF GRADE
TP	TURNING POINT/TEMP BENCHMARK
TW	TOP OF WALL ELEVATION
TYP	TYPICAL
VERT	VERTICAL
UE	UNDERGROUND ELECTRIC
UNO	UNLESS NOTED OTHERWISE
UT	UNDERGROUND TEL
W	WATER
W/	WITH
W/O	WITHOUT
WTP	WASTEWATER TREATING PLANT
XING	CROSSING

DESIGN STANDARDS

THE APPLICABLE CIVIL DESIGN STANDARDS ARE AS FOLLOWS:

UFC 2-000-05N	FACILITY PLANNING CRITERIA
UFC 3-201-01	CIVIL ENGINEERING
UFC 3-230-01	WATER STORAGE, DISTRIBUTION AND TRANSMISSION
UFC 3-240-01	WASTEWATER COLLECTION
UFC 3-250-01	PAVEMENT DESIGN FOR ROADS AND PARKING AREAS
UFC 4-022-03	FENCES AND GATES
ARCHITECTURAL BARRIERS ACT (ABA)	ACCESSIBILITY GUIDELINES

LEGEND

EXISTING		NEW
	BENCHMARK	
	FIRE HYDRANT	
	VALVE	
	FIRE DEPARTMENT CONNECTION	
	POST INDICATOR VALVE	
	LIGHT POLE	
	POWER POLE	
	GUY WIRE	
	ELECTRIC MANHOLE	
	ELECTRIC HANDHOLE	
	COMMUNICATION MANHOLE	
	COMMUNICATION HANDHOLE	
	STORM MANHOLE	
	SEWER PUMP STATION	
	OIL/WATER SEPARATOR	
	STORM DROP INLET	
	TRENCH DRAIN	
	TRANSFORMER	
	COMM LINE (UNDERGROUND)	
	FIBER OPTIC (UNDERGROUND)	
	ELEC LINE (UNDERGROUND)	
	ELEC LINE (OVERHEAD)	
	WATERLINE	
	SANITARY SEWER LINE	
	SANITARY FORCE MAIN	
	STORM DRAIN PIPE	
	FUEL LINE	
	MAJOR CONTOUR	
	MINOR CONTOUR	
	SECURITY FENCE	
	SILT FENCE	
	SPOT GRADE	
	BOLLARD	
	STORMWATER FLOW ARROW	
	STRIPING ERADICATION	
	CONCRETE SIDEWALK	
	ASPHALT PAVEMENT	
	BUILDING	
	HEAVY DUTY PORTLAND CEMENT CONCRETE	
	MECHANICAL YARD CONCRETE PAVEMENT	

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ACTIVITY

D. EVANS VIA EMAIL

SATISFACTORY TO: DATE 01/15/2026

DES	MAD	DRW	DTK	CHK	JWN
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PM/DM	MJD/ATP
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BRANCH MANAGER	JMW
SEE PAGE 202	WDE

DES PROJ DIR	WBF
FIRE PROTECTION ENGINEER	DSN

ND	WA		
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CONTINUATION

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PROJECT NO.:	1838720

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NAVFAC DRAWING NO.
14157081

SHEET 19 OF 170

C 003

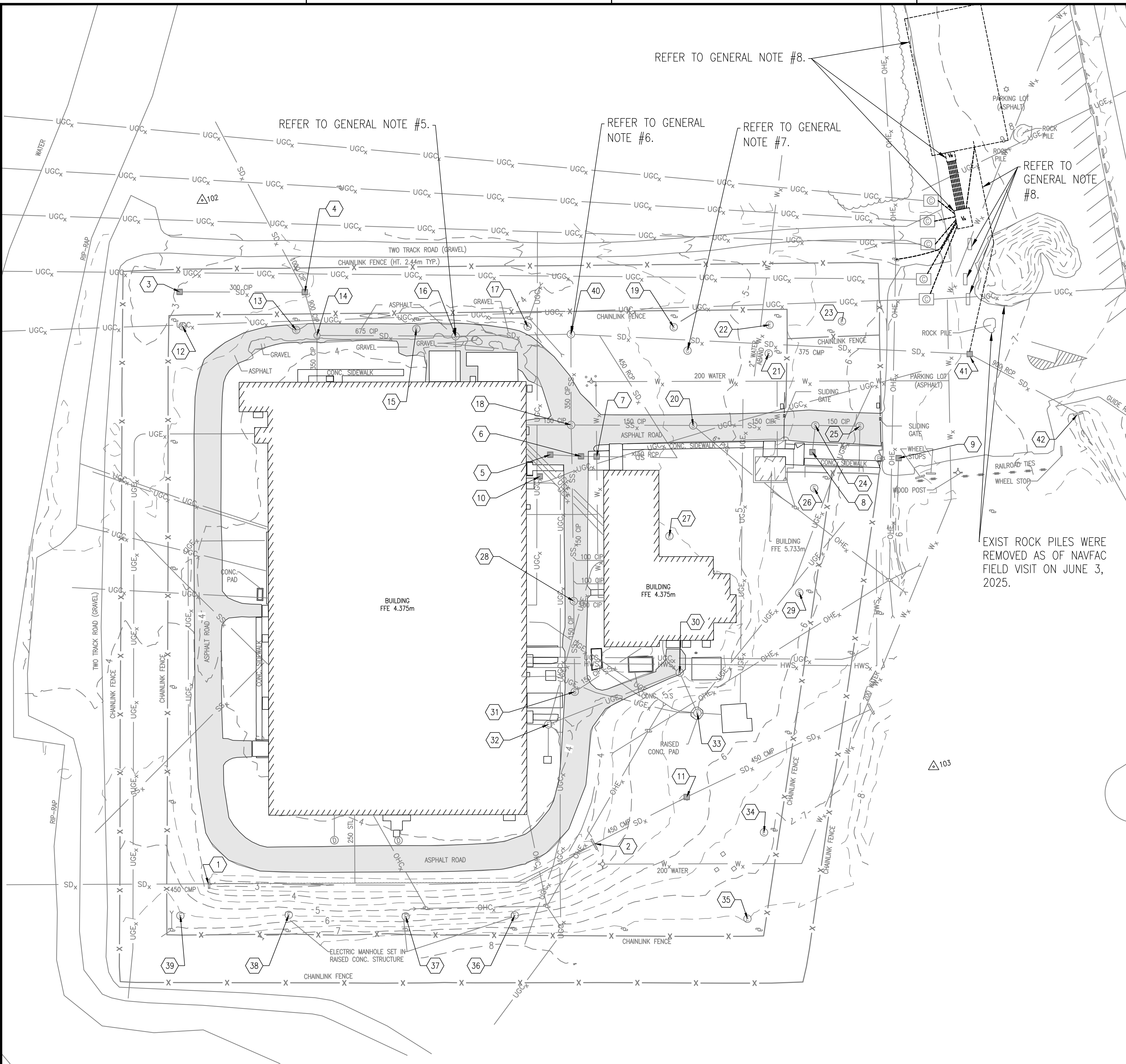
C-002

FAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: V:\DDONIS\MultipleSta\2026_1783264-P1360_MMF\B_Design\DWG\Sheet\03_SURV\1500-1783264-VF101.dwg LAYOUT NAME: VF101 PLOTTED: Tuesday, January 20, 2026 - 10:45am USER: morym.durant

EXISTING CONDITIONS PLAN

SCALE 1:400



REFER TO GENERAL NOTE #8.

REFER TO GENERAL NOTE #5.

REFER TO GENERAL NOTE #6.

REFER TO GENERAL NOTE #7.

REFER TO GENERAL NOTE #8.

EXIST ROCK PILES WERE REMOVED AS OF NAVFAC FIELD VISIT ON JUNE 3, 2025.

GENERAL NOTES

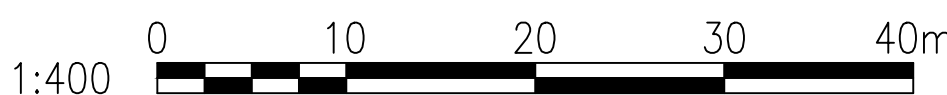
1. REFER TO SHEET C-001 FOR CIVIL LEGEND, NOTES & ABBREVIATIONS.
2. REFER TO SHEET B-101 FOR GEOTECHNICAL INFORMATION.
3. REFER TO SHEETS CE101 & CE102 FOR EROSION & SEDIMENT CONTROL REQUIREMENTS.
4. REFER TO SHEET C-001 FOR SURVEY CONTROL MONUMENT INFORMATION.
5. SURVEY BY MICHAEL BAKER WAS UNABLE TO OPEN LID. INVERT ELEVATIONS SHOWN ARE APPROX AND WERE COLLECTED FROM A FIELD VISIT BY NAVFAC LANT ON JUNE 3, 2025.
6. SURVEY BY MICHAEL BAKER WAS UNABLE TO LOCATE MANHOLE. INFORMATION SHOWN ARE APPROX AND WERE COLLECTED FROM A FIELD VISIT BY NAVFAC LANT ON JUNE 3, 2025.
7. COMM VAULT NOT LOCATED ON MICHAEL BAKER SURVEY. LOCATION SHOWN IS APPROX FROM FIELD VISIT BY NAVFAC LANT ON JUNE 3, 2025.
8. REFER TO EXWC SHORELINE IMPROVEMENT PLANS PREPARED BY ROICC-MAINE FOR ADDITIONAL INFORMATION.

STRUCTURE INFORMATION

- | | | | |
|--|---|--|--|
| 1. HEADWALL #1
TW=3.698
INV=2.858
(450MM CMP W) | 14. MANHOLE #3
SANITARY SEWER
RIM=3.893
INV=0.635
(900MM CIP N)
INV=0.915
(600MM CIP E)
INV=1.324
(350MM CIP S) | 25. MANHOLE #14
SANITARY SEWER
RIM=6.052M
UNABLE TO OPEN | 40. DOGHOUSE
SANITARY SEWER
MANHOLE BURIED
UNDER APPROX
1M OF SOIL
TOP OF PIPE (E)
=-1.2M FROM RIM
TOP OF PIPE (S)
=-1M FROM RIM
(REFER TO
GENERAL NOTE
#6 ABOVE) |
| 2. HEADWALL #2
TW=3.669
INV=3.294
(450MM CMP NE) | 15. MANHOLE #4
ELECTRIC
RIM=3.887
(300MM CIP E) | 26. MANHOLE #15
ELECTRIC
RIM=5.541M | 41. CB#8
RIM=6.544
INV=5.91
(750MM RCP SE)
INV=5.575
(350MM CMP W) |
| 3. CB#1
TG=3.019
INV=1.960
(300MM CIP E) | 16. DOGHOUSE MANHOLE #5
SANITARY SEWER
RIM=3.937
INV=1.85M
(600MM CIP E)
INV=1.96M
(350MM CIP S)
(REFER TO GENERAL
NOTE #5 ABOVE) | 27. MANHOLE #16
ELECTRIC
RIM=4.494 | 42. HEADWALL #3
TW=6.847
INV=5.819M
(750MM RCP NW) |
| 4. CB#2
TG=3.483
INV=1.724
(300MM CIP W)
INV=1.171
(900MM CIP NW)
INV=1.171
(900MM S) | 17. MANHOLE #6
ELECTRIC
RIM=3.938 | 28. MANHOLE #17
SANITARY SEWER
RIM=4.145
INV=3.152
(150MM CIP N)
INV=3.435
(100MM CIP E)
INV=3.141
(150MM CIP S) | |
| 5. CB#3
TG=4.107
INV=3.904
(350MM CIP E) | 18. MANHOLE #7
SANITARY SEWER
RIM=4.073
INV=3.130
(100MM CIP N)
INV=3.027
(150MM CIP W)
INV=2.744
(350MM CIP S) | 29. MANHOLE #18
ELECTRIC
RIM=5.662 | |
| 6. CB#4
TG=4.097
INV=3.932
(100MM CIP W)
INV=3.910
(100MM CIP E) | 19. MANHOLE #8
ELECTRIC
RIM=4.641 | 30. MANHOLE #19
ELECTRIC
RIM=4.233 | |
| 7. CB#5
TG=4.119
INV=3.829
(100MM CIP W)
INV=3.812
(80MM PVC E) | 20. MANHOLE #9
STORM SEWER
RIM=4.644
INV=3.854
(150MM CIP W)
INV=3.864
(150MM CIP E)
INV=3.909
(150MM PVC SE) | 31. MANHOLE #20
SANITARY SEWER
RIM=4.043
INV=3.363
(150MM CIP N)
INV=3.332
(150MM CIP NE) | |
| 8. CB#6
TG=5.616
INV=5.229
(300MM PVC N) | 21. MANHOLE #10
ELECTRIC
RIM=5.110M | 32. MANHOLE #21
ELECTRIC
RIM=4.037 | |
| 9. CB#7
TG=6.295
INV=5.696
(350MM CMP S) | 22. MANHOLE #11
ELECTRIC
RIM=5.253M | 33. MANHOLE #22
ELECTRIC
RIM=6.561 | |
| 10. CB#9
TG=4.147
INV=3.986
(80MM CMP NW) | 23. MANHOLE #12
ELECTRIC
RIM=5.724M | 34. MANHOLE #23
ELECTRIC
RIM=6.648 | |
| 11. CB#10
TG=5.492
INV=4.961
(300MM CMP NE)
INV=4.526
(450MM CMP SW) | 24. MANHOLE #13
SANITARY SEWER
RIM=5.686M
INV=4.630M
(150MM CIP W)
INV=4.789M
(300MM CIP S)
INV=4.645M
(150MM CIP E) | 35. MANHOLE #24
ELECTRIC
RIM=7.339 | |
| 12. MANHOLE #1
ELECTRIC
RIM=3.164 | | 36. MANHOLE #25
ELECTRIC
RIM=6.173 | |
| 13. MANHOLE #2
ELECTRIC
RIM=3.758 | | 37. MANHOLE #26
ELECTRIC
RIM=5.906 | |
| | | 38. MANHOLE #27
ELECTRIC
RIM=5.788 | |
| | | 39. MANHOLE #28
ELECTRIC
RIM=5.547 | |

PLAN
NORTH

GRAPHIC SCALE



DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
EXISTING CONDITIONS PLAN
VARIOUS LOCATIONS
NAVFAC DRAWING NO.
14157082
SHEET 20 OF 170
VF101

APPROVED: [Signature]
Win B Forbes
Date: 2026.01.28
FOR COMMANDER: NAVFAC
ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE: 01/15/2026
DES: MAD | DRW: DTK | CHK: JWN
PM/DM: MJD/ATP
BRANCH MANAGER: JMW
DES PROD DIR: WBF
FIRE PROTECTION ENGINEER: DSN



SEAL